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ReaxFF Molecular Dynamics Simulations of Hydroxylation Kinetics for Amorphous and Nano-Silica Structure, and Its Relations with Atomic Strain Energy

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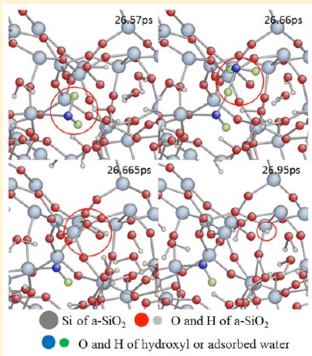
S Supporting Information

ABSTRACT: We performed reactive force field molecular dynamics simulation to observe the hydrolysis reactions between water molecules and locally strained SiO_2 geometry. We optimized the force field from J. Fogarty et al. 2010, to more accurately describe the hydroxylation reaction barrier for strained and nonstrained Si—O structures, which are about 20 and 30 kcal/mol, respectively. After optimization, energy barrier for the hydroxylation shows a good agreement with DFT data. The observation of silanol formation at the high-strain region of a silica nanorod also supports the concept that the adsorption of water molecule: hydroxyl formation favors the geometry with higher strain energy. In addition, we found three distinct hydroxylation paths—H_3O^+ formation reaction from the adsorbed water, proton donation from H_3O^+, and the direct dissociation of the adsorbed water molecule. Because water molecules and their hydrogen bond network behave differently with respect to temperature ranges, silanol formation is also affected by the temperature. The formation of surface hydroxyl in an amorphous silica double slit displays a similar tendency: SiOH formation prefers high-strain sites. Silanol formation related with H_3O^+ formation and dissociation is observed in hydroxylation of amorphous SiO_2, similar to the results from silica nano wire simulation. These results are particularly relevant to the tribological characteristics of surfaces, enabling the prediction of the attachment site of the lubrication film on silica surfaces with a locally strained geometry.



1. INTRODUCTION

Since amorphous silica is one of the most abundant materials on the surface of the planet Earth, many scientists and engineers studied silica and silicon related system. From the earliest days of history to the modern technologic era, silica has been an important material for various applications, such as glass, wafer, electronic devices, and nanotech devices.[1−4]

In addition, among all the studies related with silicon, silica−water interaction has been a central research topic, due to its importance and influence on theoretical science and practical applications. More importantly, the silica/water combination is very inexpensive and ubiquitous, which is why silica−water has often been preferred by researchers compared to other chemical combinations present in the natural environment.[5,6] Many studies have been performed to understand the interfacial surface chemistry regarding water−silica system, such as how surface silicon-hydroxyls modulate interfacial properties, or how this chemistry is affected by surface characteristics,[7,8] or by the effect of tensile or compressive stress on reaction between silica surface and water.[9]

The hydrolysis reaction between water and the silica surface is initiated by physisorption of the water molecule onto the surface, followed by its dissociation and chemisorption as a hydroxyl group. Hydroxyls on the surface are known to greatly modify surface properties, due to their unique ability to form hydrogen bond networks. It has been observed that under-coordinated surface atoms and strained sites are responsible for structural deformation, such as crack formation, corrosion, or other surface chemistry. Similarly, hydrolysis reaction also favors defects and holes on the surface.[10]

A significant research effort has been executed to understand and utilize silanols and Si—OH-related surface activity. Recent experimental techniques, such as XANES, are capable of in situ measuring of electronic structure at the silica-liquid interface.[11] NMR and IR spectroscopy on silica surface expanded the understanding of various types of silanols.[12] However, experimental approaches are often not appropriate to represent surface topography and surface structure information in the atomic size scale, and on the picosecond time-scale connected with chemical reactions.

Computational chemistry methods enable us to simulate and visualize the processes of chemical reactions as well as track the behavior of atoms and molecules during reaction. Ab initio simulations have been applied to compute the water−silica

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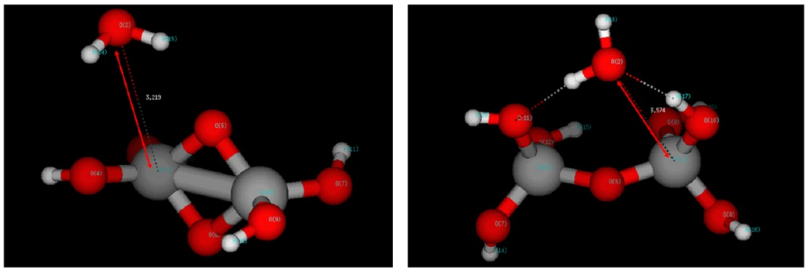

Figure 1. Picture of strained Si dimer (left) and nonstrained silica dimer (right).

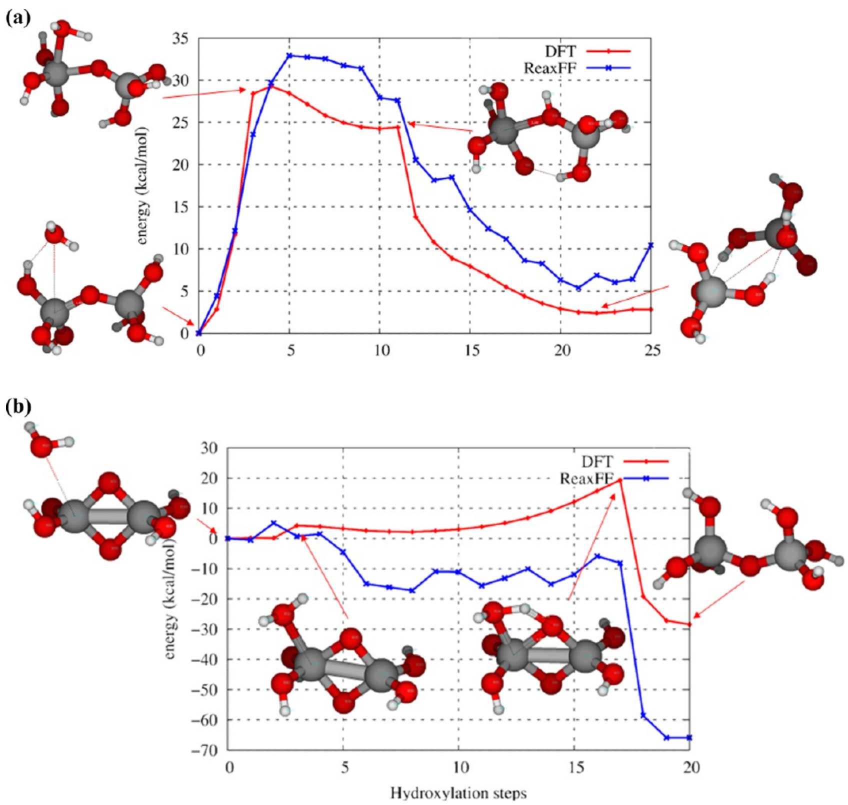

Figure 2. (a) Energy barrier for hydroxylation on nonstrained Si dimer structure with the Fogarty force field. (b) Energy barrier for hydroxylation on strained Si dimer structure with the Fogarty force field.[1]

interactions by researchers, providing the clues for understanding of interfacial nature in atomic scale. For example, Y. He et al.[9] showed the relationship between external stress and long-range surface water interaction, and role of other water molecules during reaction. Also, they predicted the predissociation adsorption of water on silica surface using a Born–Oppenheimer model. C. Mischler et al.[13] studied silica–liquid interaction using a mixed classical MD and ab initio code, with BKS and Car–Parrinello potential, to simulate reaction on the surface. M. H. Du et al.[14] also applied a DFT-assisted classic potential MD code; they revealed various trends of hydrogen bond network between crystal phase and amorphous phase of silica. They also pointed out the effect of system size on hydration energy. Y. Ma et al.[15] investigated how undercoordinated Si atoms and third-party water molecules around OH group lead the dissociation of adsorbed H_2O molecule. F. Musso et al.[16] indicated that silica–water interaction strengthens hydrogen bonding between surface hydroxyls and water molecules. However, high computational cost and limit of system size are a significant drawback for DFT and other first principle based approaches.

Empirical force field methods enable us to study silica–water interfacial science and reaction mechanisms in a larger scale, because it can save a significant amount of the computational cost, when compared to quantum mechanics simulations.[17−20] T. S. Mahadevan et al.[21] developed a dissociative water potential, and represented the silanol formation on water–silica interface. G. K. Lockwood and S. H. Garofalini[22] group's recent study using dissociative potential for water molecules on the silicon oxide showed active proton transfer among water molecules and silica surface with H_3O^+ formation. J. Fogarty et al.[1] proposed silica/water interaction properties using the ReaxFF reactive potential molecular dynamics.

Following Garofalini and Fogarty's observations, in this research, MD simulation using the ReaxFF potential for water–amorphous silica system was performed, which represented the effect of strain energy on hydroxylation. Specially, strained silicon and oxygen atoms on the surface were closely analyzed, as well as undercoordinated atoms and nonbonding-oxygen.

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We studied a SiO_2 based nanotube structure to check the force field in small, highly strained geometry, and how water molecule interacts with and reacts to the strained site of this geometry. Thereafter, we investigated the surface chemistry of large amorphous silica structures.

Previous computational approaches with reactive potential predicted the energy relation for hydroxylation, and role of hydronium.[21,22] In this study, using ReaxFF potential, first, we reparametrized the Si—O-water interaction parameters, in order to accurately simulate the energy barrier of hydroxylation to silicon. Second, with the advanced force field, ReaxFF MD simulation was calculated to reveal the relationship between atomic strain energy and hydroxylation, with smaller silica nano wire, and with amorphous SiO_2 slab. In addition, after series of hydroxylation MD simulation under various temperature condition, and importance of 2−3 paired water dimer/polymers connection was observed.

To discuss details, section 1 explains the validation and optimization of force field using two Si dimer geometries (Figure 1). Section 2 introduces the silica nano wire (Figures 2−4), and observation of hydroxylation on silica nano wire, which prefers the strained edge sites (Figure 5). In section 3, representative examples for three distinctive reaction pathways around strained sites are shown: (i) hydroxylation involving formation of H_3O^+ (Figure 6), (ii) hydroxylation involving dissociation of H_3O^+ (Figure 7), and (iii) hydroxylation involving dissociation of adsorbed water molecule (Figure 8).

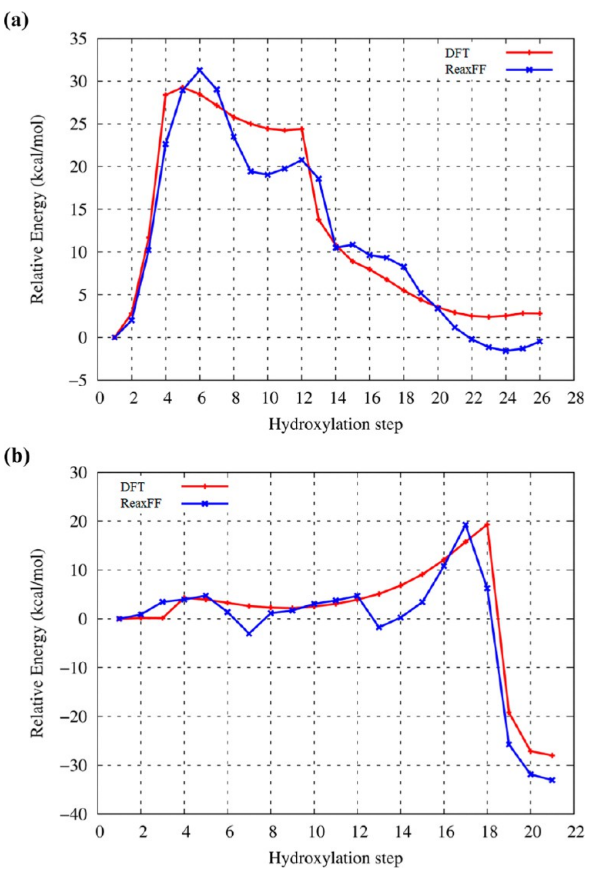

Figure 3. (a) Energy barrier for hydroxylation of the nonstrained silica dimer structure for the SiO-2015 force field. (b) Energy barrier for hydroxylation of the strained silica dimer structure for the SiO-2015 force field.

Section 4 discusses the analysis of number of hydroxylation (Figure 9), RDF of water−Si (Figure 10), and analysis of water clusters (Tables 1 and 2, and Figure 11), revealed the effect of water cluster on hydrolysis reaction and hydronium formation. Lastly, section 5 describes the effect of locally strained structures on amorphous SiO_2 slab on hydroxylation, showing silanol formation favored the surface atoms with high strain energy (Figures 12, 13, and 14). Reaction pathways for the hydroxylation on slab surface are also shown. (Figures 15 and 16). All visualizations were performed by Molden, Atomeye, and Ovito.[23]

2. COMPUTATIONAL DETAILS—REACTIVE FORCE FIELD: METHOD AND OPTIMIZATION

The ReaxFF Reactive force field has been developed to simulate large system ($>10^6$ atoms) with the relatively small computational cost, with a comparable accuracy to quantum based simulation results.[24]

ReaxFF employs a bond-order dependent concept, allowing formation and breaking of bonds.[24] In order to calculate the contributions of these energies, all bond orders are calculated and modified at every iteration, thus allowing ReaxFF to form a relationship between the bond distance and the bond order.[25−27] ReaxFF calculates nonbonded interactions among all atom pairs. Also, ReaxFF uses an electronegativity equalization model to calculate charges and polarization effects.[28,29] Initially, a ReaxFF potential was developed to simulate hydrocarbon based system,[24] and later, the method was developed further for metals,[30] energy and electro chemistry,[31] etching and surface reaction,[32] stress analysis of material,[33] and graphene,[34] for example.

The quality of the parameters in the force field is critical, since it determines the overall accuracy of result of simulations. It is essential to check and validate the credibility of the force field before we start any meaningful simulation by training and validating against DFT or any level of quantum mechanics calculation. If the force field fails to predict the expected value, then the parameters of the force field should be refitted with experimental and QM results using a training set. Typically, this training set contains bond dissociation/angle/torsion energies, equation of state, heat of formation and energy relations for important species, reaction barrier and reaction pathway, and transition state results.

The parameters of the force fields are optimized with a single-parameter search algorithm.[35] The error function for the training set, which minimizes the error values between ReaxFF value and QM value, is shown in eq 1

$$\text{error} = \sum_{i}^{n} \left(\frac{X_{i,\mathrm{QM}} - X_{i,\mathrm{ReaxFF}}}{\sigma_{\mathrm{i}}} \right)^2, \quad i = 1, 2, 3 \cdots n \tag{1}$$

where the $X_{i,\mathrm{QM}}$ is any energy or values from experiment/quantum mechanics software, while the $X_{i,\mathrm{ReaxFF}}$ is the values from ReaxFF calculation, σ_{i} is the weight value assigned to each data point. Experimental or quantum mechanics data for force field training are usually gathered from literature and papers, or obtained from research collaborators, or calculated by using QM software. Typically, well trained parameters of ReaxFF should reach the comparable level of energy values when compared to QM or experimental values, within 2 kcal/mol for important heat of formation or energy curves/relations, and 5 kcal/mol for general reaction barriers.

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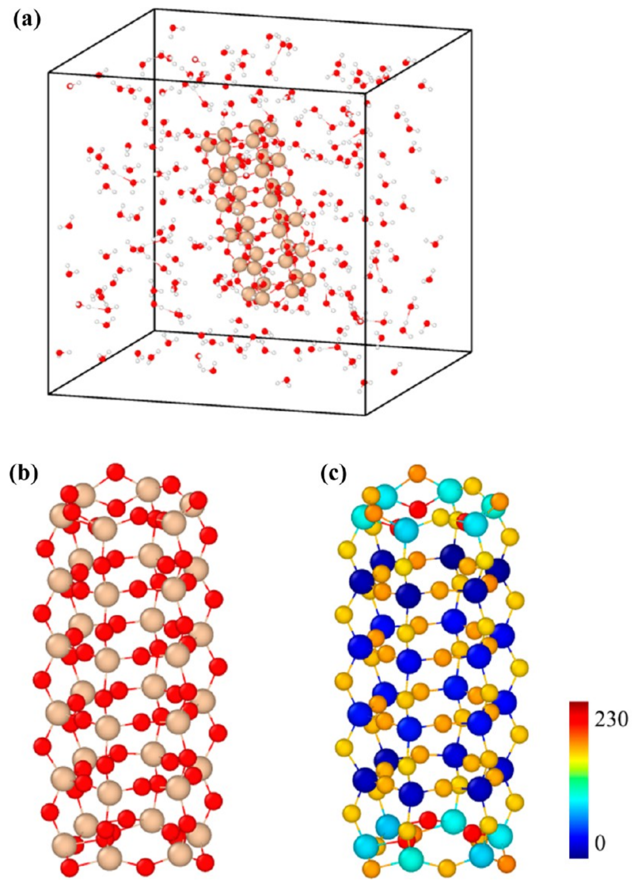

Figure 4. (a) Silica nano rod in the 200 water molecules in the 30 × 30 × 30 Å^3 periodic box. Water density was 0.222 kg/dm^3. (b) Silica nano wire. Gray atoms are Si atoms, and Red atoms are Oxygens. (c) Relative strain energy (kcal/mol) view of silica nanowire. Red atoms represent higher strained atoms, blue atoms mean lower strain energy ones.

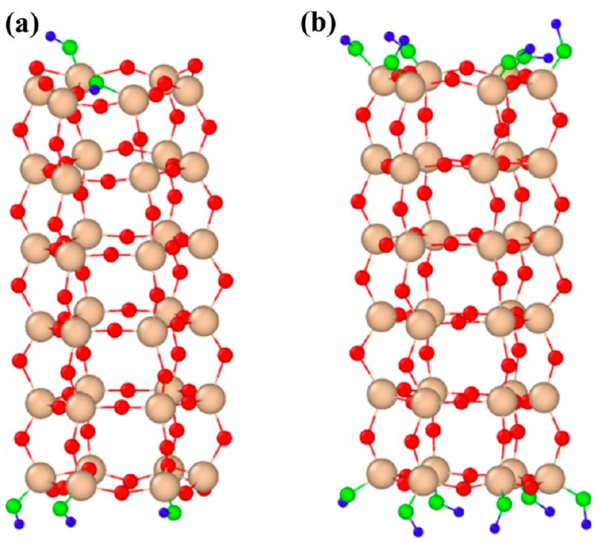

Figure 5. (a) Silica nanowire after 650 ps exposure at 400 K temperature. Five silanols are formed at the edge area. Green and blue atoms represents the O and H of the silanol. (b) Silica nanowire, 500 ps exposure at 700 K temperature. All strained Si atoms at the edge sites are occupied by silanols.

2.1. Validation and Optimization of the ReaxFF Force Field for Hydroxylation Reactions in Silica. Using DFT and ReaxFF, energy scan for the transition state was performed in order to examine and compare how the reaction energy barrier for silanol formation varies with strained geometric conditions. As shown in Figure 1, we prepared two geometries of a silica dimer: One has the 4-membered Si—O—Si—O ring structure with a high strained energy, and the other has only one bridge oxygen with relatively low strain energy. All DFT calculations were carried out using B3LYP functional with the 6-311++G(d,p) basis set. To calculate the energy barrier using DFT, first we calculated transient state for the hydroxylation, and applied internal coordinate search to check reaction pathways. Then the bond distance scan was applied to get energies for each points. ReaxFF potential developed by J. Fogarty et al.[1] was applied to check the ReaxFF hydroxylation energy barrier. As a result, energy barriers calculated by previous Si—O-water interaction parameters[1] were shown to be slightly insufficient when compared to DFT result. Figure 2(a and b) presents the energy curves as a water molecule approaches one of Si atoms of the strained and nonstrained silica dimer structure, respectively. Both hydroxylation reactions are initiated by adsorption of water molecule to a Si atom of dimer, and proton donation occurs to the bridge oxygen. Then, one of the Si—O bonds breaks, forming a system with 2 silanol groups. As we can see in Figure 2(a), the energy curve for a nonstrained silica dimer correctly followed the energy curve from DFT. However, for strained Si—O ring structure, the energy barrier in Figure 2(b) for hydroxylation was underestimated when compared to that of DFT, indicating easier hydroxylation for strained structure, for the force field for Si—O-water interaction from J. Fogarty et al.[1]

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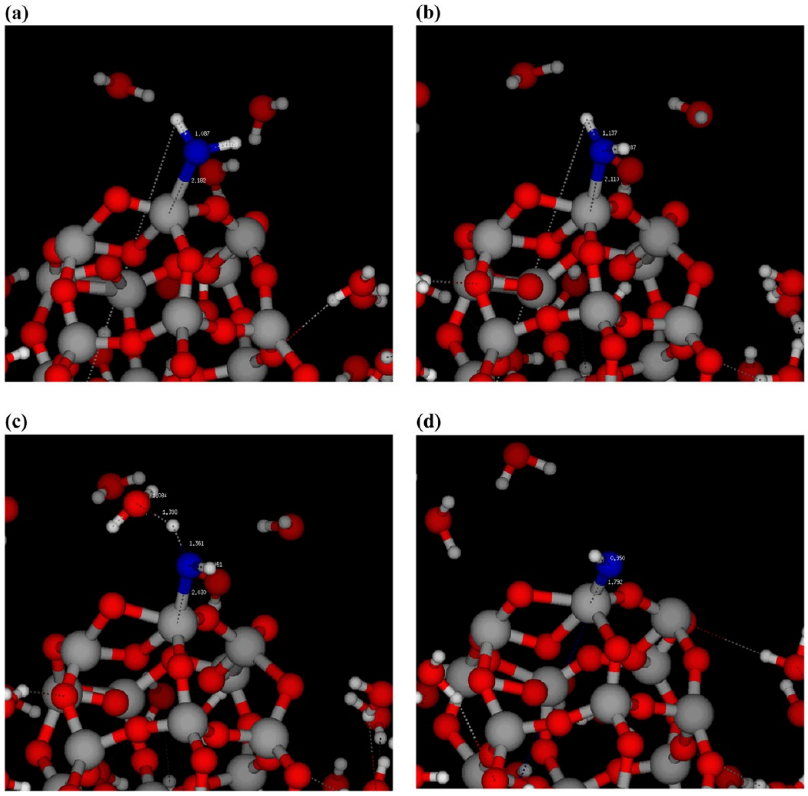

Figure 6. Snapshots of hydroxylation reaction, involving the formation of hydronium ion at 400 K. Si, O, and O of adsorbed water molecule or hydroxyl, H. (a) 76.21 ps after MD simulation. Adsorbed water molecule at the edge Si of silica nano wire. (b) 76.26 ps after MD simulation. One of the H atom of adsorbed water molecule is attracted to nearby secondary water molecule. (c) 76.265 ps after MD simulation. Proton donation occurred. (d) 76.32 ps after MD simulation. Formation of hydroxyl. H_3O^+ ion moved across the periodic boundary.

In order to improve the ReaxFF performance for the strained silica dimer, we added the two DFT energy curves for hydroxylation on the strained silica dimer, and nonstrained silica dimer to the Fogarty et al. training set,[1] and reparameterized the force field, generating the SiO-2015 parameter set. Trained parameters are Si—O bond and off-diagonal parameters, Si—O—Si, O—Si—O, O—Si—Si, and H—O—Si angle parameters, and O—H—O hydrogen bond parameters. The energy curve result for two hydroxylation reaction after force field optimization is shown in Figure 3(a and b). After parametrization, the hydroxylation energy barrier of new force field is about 31 kcal/mol for nonstrained Si dimers, and 20 kcal/mol for strained Si dimer. Note that the data in Figure 3(a and b) are not corrected for zero-point energies (ZPE), but we evaluated ZPE for both products and reactants, and found that this correction is less than 1 kcal/mol.

2.2. Application to Silica Nano Wire Hydrolysis. As a test of the SiO-2015 force field, a SiO_2 nanowire and 200 water molecules were simulated to check the relationship between strain energy and formation of silanol in a smaller and simpler system. The silica nanowire structure of this research has also been used in previous studies.[36,37] System configuration is shown in Figure 4(a). Figure 4(b) is the picture of silica nano wire, and Figure 4(c) presents the strain energy of each atom in silica nano wire.

The silica nano wire in our study, which was used in previous research to find the dependence of silica−water reaction,[37] has distinctively strained region at the edge of structure. As seen in Figure 4(b and c), the silica nanowire, which is nonperiodic and hollow, consists of 36 silicon and 72 oxygen atoms. This nanowire's top and bottom surfaces are opened to vacuum, thus creating a highly strained, Si—O—Si-O 4 membered ring structure at the edge. Figure 4(c) clearly shows the strained Si and O atoms at the edge area, represented by different colors with a relatively higher strain energy, while the atoms in the body are less strained. There are total 12 of Si atoms at the edge of the silica nano wire, which contains higher relative strain energy than other Si atoms do. Average relative strain energy of Si atoms in nonstrained structure was 15.97 kcal/mol, while that of Si atoms in strained structure was 78.09 kcal/mol. The orthogonal periodic box has a volume of $30 \times 30 \times 30$ Å^3, and the density of water equals to 0.22 kg/dm^3. 800 ps NVT MD simulations were performed by applying the Berendsen thermostat to all atoms in the system, with 100 fs of temperature damping constant, and 0.1 fs of time step. System temperature was set from 300 K to 1500 K, and simulations were performed at every 100 K. The reason behind the choice of wide temperature range was, first, to check the behavior of water clusters as environment approaches the supercritical state of water vapor, and second, to reveal how the hydroxylation is influenced by high temperature. The simulations were performed at 100 K temperature intervals, to distinguish the effect of different temperatures.

Initial observation of MD simulation showed that the hydroxyl formation was initiated from the edge area. Figure 5(a and b) shows the pictures of silica nano wire and Silanols formed at the edge of the Silica nano wire, in the 400 K at 650 ps and 700 K at 500 ps, respectively. Clearly, hydroxyl formation occur preferably at the strained Si—O sites of silica nano wire, which implies that the approaching and adsorption

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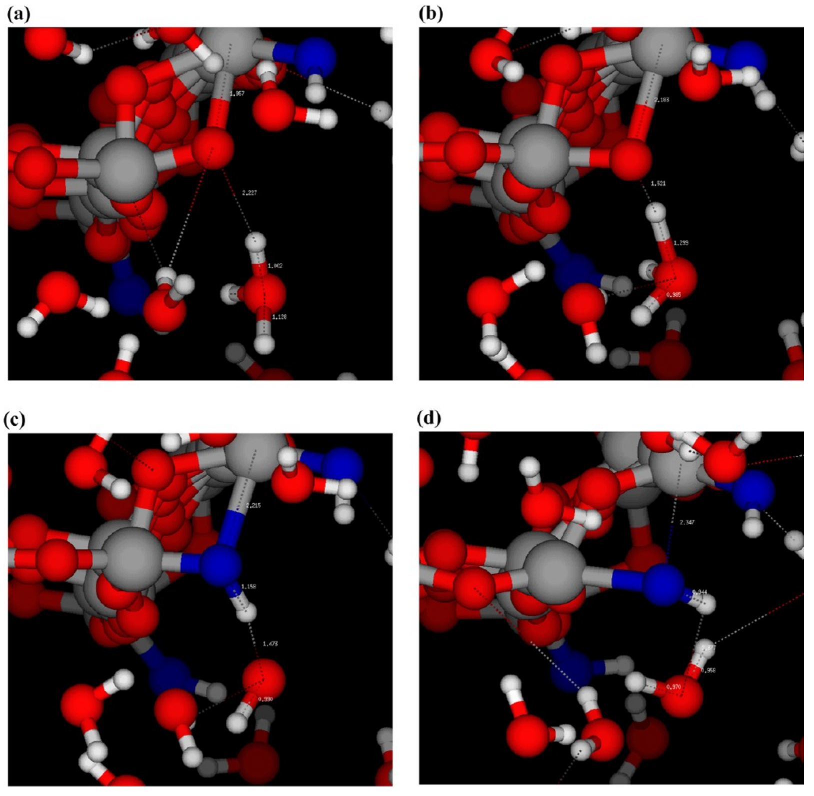

Figure 7. Snapshots of hydroxylation reaction, involving the dissociation of hydronium ion at 900 K. All color settings of atoms are the same as Figure 6. (a) 195.3 ps after MD simulation. H_3O^+ approaches to bridge O. (b) 195.37 ps after MD simulation. One of the H of H_3O^+ is attracted to the bridge O of strained structure. (c) 195.375 ps after MD simulation. Proton donation. (d) 195.45 ps after MD simulation. Si—O bond breaks, and SiOH formed.

of the water molecule also prefer the strained edge site of the silica nano wire.

2.3. Reaction Path for Hydroxylation on the Edge Area of Silica Nano Wire.

As G. K. Lockwood et al.[22] observed in a previous study using a dissociative potential, the hydronium ion plays an important role in the silanol formation reactions. Figure 6 describes the process of hydroxylation at 400 K, which includes the formation of a H_3O^+ ion. At 76.21 ps (Figure 6(a)), a water molecule adsorbed to the Si atom of the edge of silica nano wire. At 76.26 ps (Figure 6(b)), a secondary water molecule closely approached the adsorbed water molecule. At 76.265 ps (Figure 6(c)), proton donation was occurred. Finally, at 76.32 ps (Figure 6(d)), a H_3O^+ and a silanol were formed. The hydronium ion, generated in Figure 6(d), moved away from the hydroxyl group, forming a hydrogen bond network with other nearby water molecules and hydroxyls.

In addition, dissociation of hydronium ion was also a major route for silanol formation. Figure 7(a−d) show an example of H_3O^+ deprotonation path at 900 K condition. At 195.3 ps (Figure 7(a)), a H_3O^+ approaches Si—O—Si—O ring structure at the edge of the silica nano wire. At t = 195.37 ps (Figure 7(b)), one of the H^+ ions of hydronium interacts with bridge oxygen of silica nano wire, which has a relatively high strain energy. At 195.375 ps (Figure 7(c)), H_3O^+ deprotonates one of H^+ to the O and becomes H_2O. Lastly, at 195.45 ps (Figure 7(d)), one of the Si—O bonds is elongated, forming a hydroxyl and H_2O molecule as a product.

Not only hydronium ions, but also water molecules can induced the hydroxylation reaction, especially at higher temperature, (typically T > 1000K). Figure 11(a)−(c) presents an example of the formation of silanol due to the direct dissociation of an adsorbed water molecule, which tends to be observed at higher temperatures in our simulations. At 121.76 ps (Figure 8(a)), a water molecule was adsorbed on the Si atom at the edge area of silica nano wire. At 121.815 ps (Figure 8(b)), one of the H atoms at the adsorbed water approaches the bridge oxygen, stretching O—H distance to 1.1∼1.2 Å. At 121.82 ps (Figure 8(c)), after H moves to the bridge oxygen, Si—O distance increases to more than 2 Å. At 121.87 ps after simulation (Figure 8(d)), 2 hydroxyls were formed as one of the Si—O bonds dissociated.

2.4. Analysis of Hydroxyl Formation with Respect to Different Temperatures.

Our observations from the reaction pathways of hydroxylation imply that there is a relationship among temperature, water clusters, and the kinetics of hydroxyl formation. Accordingly, an analysis of the reaction path for hydroxylation was performed to reveal the connection among them. Figure 9 shows the maximum number of silanols during each simulation. According to Figure 9, the maximum number of silanols during the simulations is not proportional to the temperature. Instead, the number of silanols shows the maximum number of 14 at 900 K, 1000 K, and 1100 K, and minimum number of 5 at 400 K. In 300 K simulation, no silanol formation is observed. On the basis of our observation of reaction kinetics from 900 K to 1100 K, 2 hydroxyls—13th and 14th—are generated at the Si atom of body area of silica nano wire by proton donation from H_3O^+, only after all 12 strained Si atoms at the edge of the silica nano wire are fully hydroxylated. Time tracking of the number of silanols is shown in the Supporting Information (Figure S1), which

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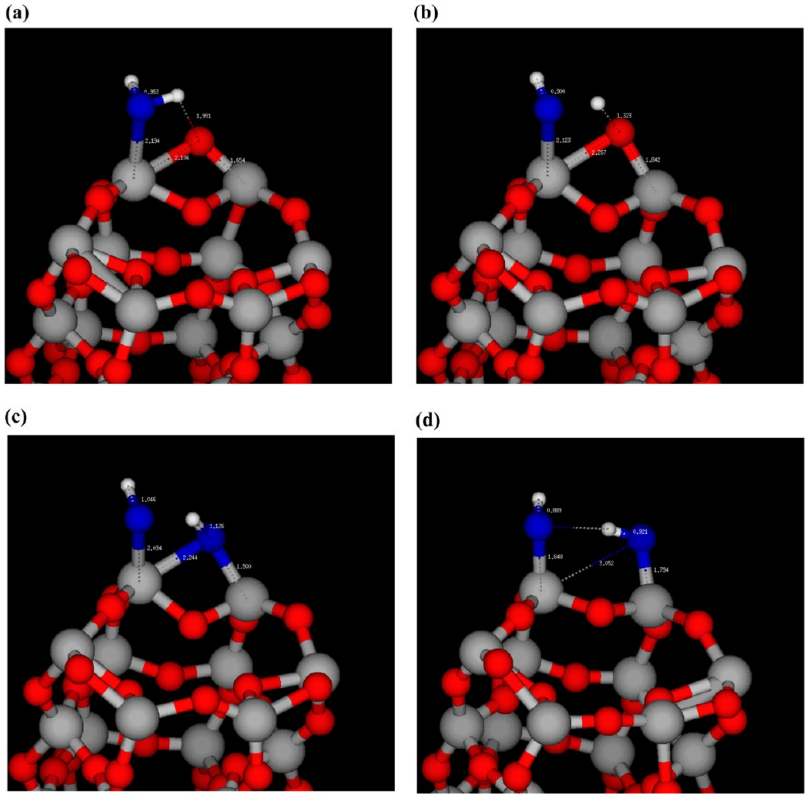

Figure 8. Snapshots of hydroxylation reaction, involving the dissociation of adsorbed water molecule at 1500 K. All color settings of atoms are the same as Figure 6. Water molecules, unrelated with reaction, are removed for clarity. (a) 121.76 ps after MD simulation. Adsorption of water to Si. (b) 121.815 ps after MD simulation. One of the H of adsorbed H_2O is attracted to the bridge O of strained structure. (c) 121.82 ps after MD simulation. Proton donation. (d) 121.87 ps after MD simulation. Si—O bond breaks, and two silanols formed.

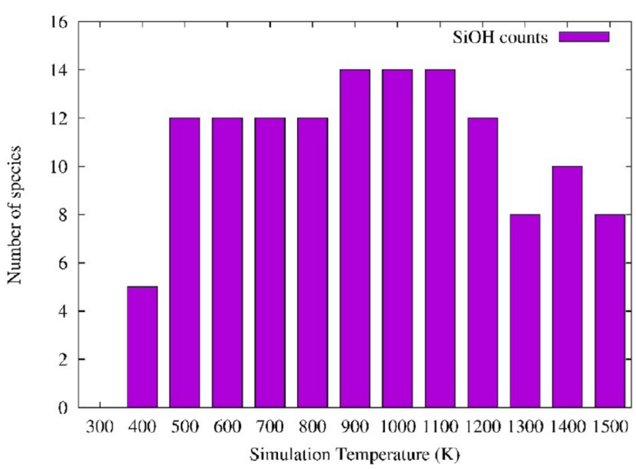

Figure 9. Maximum number of silanols during simulation for each temperature after 800 ps MD simulation.

explains the effect of the temperature on the speed of hydroxylation.

To check how water molecules behave and interact with silica nano wire under various temperature conditions, the RDF between water and Si, and RDF among waters were calculated. Water–Si RDF was calculated from the RDF of O_{water}—$Si_{nanowire}$, which is shown in Figure 10(a–c). Figure 10(d) shows the RDF among the water molecules.

According to the first peak of O_{water}—$Si_{nanowire}$ RDF data, (positioned at R_{Si-O} around 1.55 Å) at 300 K and 400 K (Figure 10(a)), water molecules hardly approach the Si atoms, resulting in no or a few water adsorption. Accordingly, the chance for the formation of silanols is lower than that in higher temperature condition. However, from 500 K to 1100 K (Figure 10(a and b)) a significant increase of the first peak of RDF is observed, which indicates a higher number of approaching of water molecules to the Si atoms. In addition, as temperature increases higher than 1200 K (Figure 10(c)), the first peak of O_{water}—$Si_{nanowire}$ RDF decreases and overall curve is flattened. RDF among water molecules (RDF of O_{water}—O_{water}, Figure 10(d)) implies similar trends with the RDF of O_{water}—$Si_{nanowire}$. 300 K water RDF presents the highest first peak, and existence of second and third peak, which suggests the strong hydrogen bond network among the water molecules. From 500 K to 1100 K of temperature, hydrogen bond network is weakened, as well as all peaks of RDF curves. After 1200 K, because of the high velocity of the water molecules, all hydrogen bond networks are broken, and the curves are greatly flattened. These RDF data also support the existence of optimal temperature ranges for the silanol formation, which was suggested in Figure 9.

However, still, it is unclear how the reaction pathways of hydroxylation are related with temperature. To find this, we investigated how hydroxyls were formed with each temperature condition. Table 1 presents how silanols were formed during all simulations. Our observation indicated that all of reaction pathways generally followed the examples shown in Figures 6, 7, and 8. According to these data, the dominant pathways for silanol formation are strongly affected by temperature. Among three methods of hydroxylation, the majority of them originate from the reaction regarding the hydronium ion. Especially,

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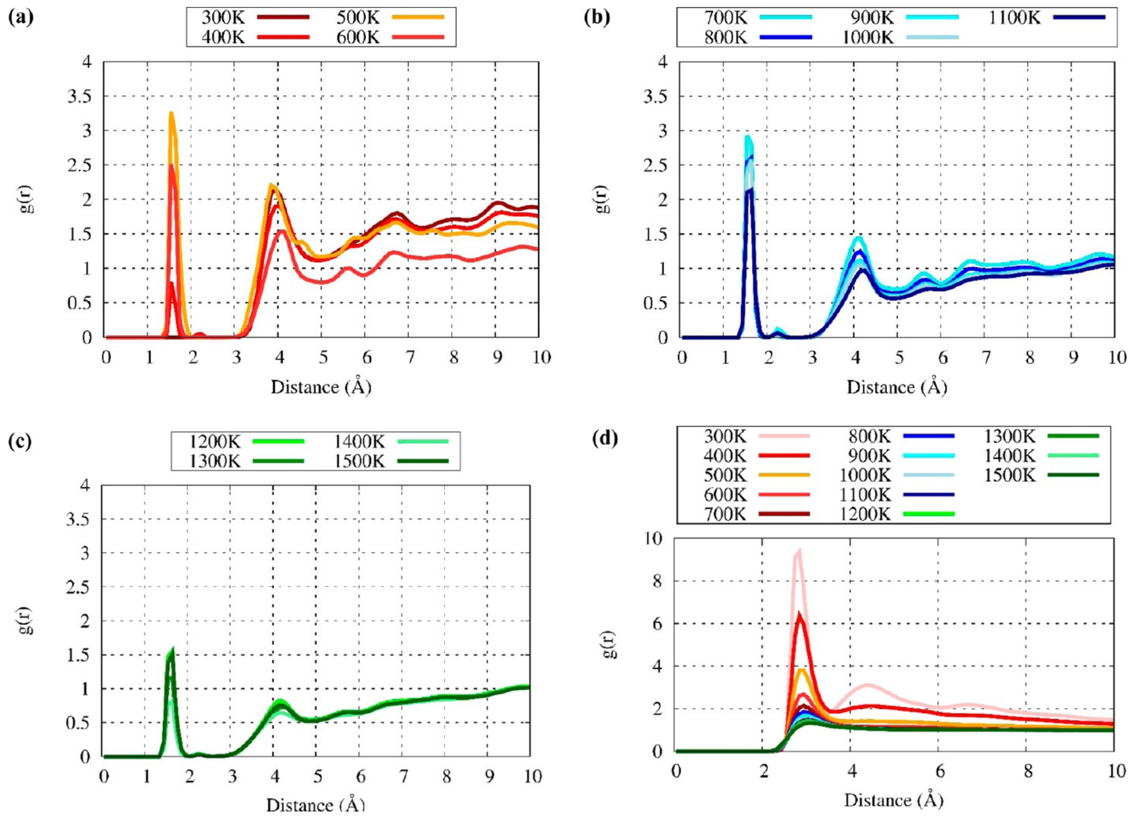

Figure 10. (a) Si—O$_{water}$ RDF at 300, 400, 500, and 600 K. (b) Si—O$_{water}$ RDF from 700 K to 1000 K. (c) Si—O$_{water}$ RDF from 1200 K to 1500 K. (d) RDF among water molecules.

Table 1. Different Types of Silanol Formation Reaction for Different Temperature

temperature (K)	total	H_3O^+ formation reaction	deprotonation from H_3O^+	H_2O dissociation
300	0	0	0	0
400	5	3 (60%)	2 (40%)	0
500	12	6 (50%)	6 (50%)	0
600	12	6 (50%)	6 (50%)	0
700	12	6 (50%)	6 (50%)	0
800	12	6 (50%)	6 (50%)	0
900[a]	14	7 (50%)	7 (50%)	0
1000[a]	14	7 (50%)	7 (50%)	0
1100[a]	14	7 (50%)	5 (36%)	2 (14%)
1200	12	6 (50%)	4 (33%)	2 (17%)
1300	8	3 (38%)	3 (38%)	2 (24%)
1400	10	4 (40%)	4 (40%)	2 (20%)
1500[b]	8	3 (38%)	2 (25%)	4 (37%)

[a]From 900 K to 1100 K, silanol formation in the body area of silica nano wire was observed, by proton donation from H_3O^+ ion. [b]At 1500 K, one of the silanol was restored to the H_2O, by H_3O^+ + SiOH → $2H_2O$ reaction.

Table 2. Average Number of Independent Water and Weakly Bonded Water Clusters[a]

temperature	isolated water	2 water	3 water	>3 water cluster	mass of >3 water cluster (AMU)
300	17.8	3.4		9.7	3397.2
400	54.8	12.5	5.2	17.8	2101.5
500	95.7	17.4	5.8	17.0	1084.6
600	124.9	18.4	4.6	13.6	593.6
700	137.7	16.9		10.4	527.1
800	146.2	15.7	2.8	6.0	270.0
900	151.8	14.6	2.3	6.2	255.4
1000	154.6	14.1		7.6	334.9
1100	157.8	13.7		7.7	326.3
1200	159.4	13.3		8.2	334.2
1300	162.1	13.4		11.9	383.8
1400	163.1	13.3		12.3	392.6
1500	163.4	12.8		13.3	405.9

[a]Larger cluster ranges from 4 water to 20 water molecules. The mass of larger cluster is an averaged value.

from 300 K to 1000 K of temperature ranges, all of the silanols are closely related with the formation and dissociation of the H_3O^+ ion. On the other hand, silanol formation due to the direct dissociation of a water molecule is observed only from 1100 K and higher temperature range. In addition, its ratio becomes the maximum at the 1500 K, the highest temperature in our study. This unimolecular silanol formation path is due to strong vibration of O—H bond of adsorbed water molecules, facilitated by high temperature environment. Because of its rapid vibrational motion, once being adsorbed by silica nano wire, water molecules can donate their H atom to the bridge oxygen of silica nano wire more easily, compared to a lower temperature environment.

To answer why the hydrolysis reaction pathways differ with respect to temperature, we analyzed the cluster of water molecules: how many water molecules did establish hydrogen bond interactions? Table 2 contains the average number of isolated water, water clusters that consisted of two or three water molecules, and the larger clusters with many water molecules interacting in hydrogen bond networks. Using the

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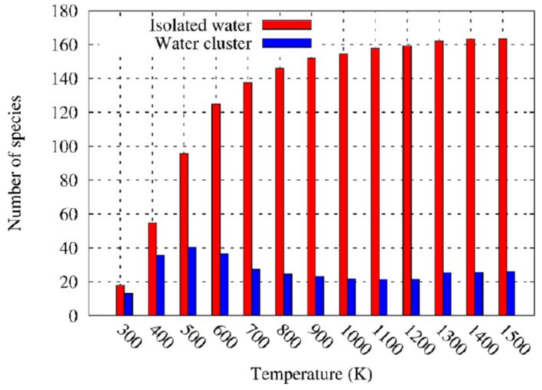

Figure 11. Average number of isolated water molecules and water clusters.

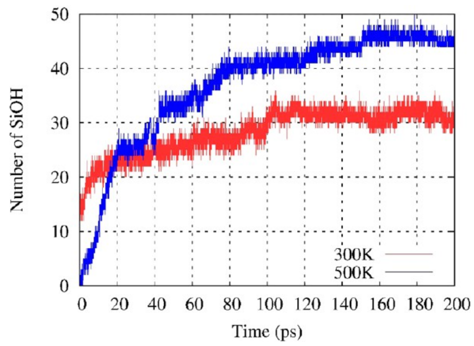

Figure 12. Number of silanols on the slab surface during hydroxylation simulation at different temperatures.

xyz trajectories of water molecules and silica nano wire from each 800 ps simulations, we run the single point MD simulations with a low bond order cutoff value of 0.1. This enables us to distinguish the hydrogen bond networks among water molecules from other chemical connections. Figure 11 shows the number of isolated water molecules, and number of water clusters. A water clusters consist of at least two water molecules.

Observation from water cluster analysis (Table 2 and Figure 11) implies the existence of relationship between the hydroxylation mechanism, temperature, and water cluster. First, water loses its autocatalytic reaction at high temperature (>1200 K), since the hydrogen bonding connections of water cluster are disrupted. Accordingly, the number of water dimers declined as temperature increases after 600 K (Table 2), and the number of hydroxyls also declined (Figure 9). This influences the pathway for hydrolysis reaction, as the ratio of dissociation of adsorbed water increased from 1100 K (Table 1). On the basis of our observations of reaction path, (Figures 6 and 7) hydrolysis reaction involving formation or dissociation of hydroxyl requires water dimers and neighboring waters to donate or to receive proton ions. In this case, the absence of water dimer structure induced the lack of hydroniums, which leads the increased number of hydroxylation by dissociation of water. This result is also presented in RDF data, showing the absence of second and third peak in water RDF curves at temperature greater than 1100 K (Figure 10(d)), and lower first peak intensity at O_{water}—$Si_{nanowire}$ RDF (Figure 10(c)). This RDF results also indicate the decreased number of water dimer structure at corresponding temperature range.

Water dimer and large water cluster structures can survive and sustain their H-bond connections in the lower temperature range. According to the data from Table 2, at the temperature range from 500 K to 1000 K, the average number of water dimer structures was larger than that at either lower or higher temperatures. This implies that water molecules in such temperature range established more sustainable hydrogen bond network, which offered stable supply of neighboring water molecules to donate or receive the proton. This results increased chance of hydrolysis reaction involving hydronium ions, as well as greater number of hydroxyls than that at extreme low- or high-temperature range (T < 500 K or T > 1300 K range) RDF curves for O_{water}—$Si_{nanowire}$ for temperature from 500 K to 1000 K (Figure 10(a, b) show a strong second peak around 4 Å, ($g(r) > 1$ at $R_{Si-O} \approx 4.1$ Å), which suggests strong hydrogen bond network and neighboring water molecules. Meanwhile, the second peak of O_{water}—$Si_{nanowire}$ for high temperature (T > 1200 K) shows weaker intensity ($g(r) < 1$ at $R_{Si-O} \approx 4.2$ Å), which means the dissociation of hydrogen bond networks, and water cluster connections.

In addition, this observation may imply the possible existence of the optimum temperature range for the hydrolysis reaction, where the reactivity of water with silica structures is maximized at such temperature.

2.5. Hydroxyl Formation on Amorphous Silica Surface: ReaxFF MD Simulation Results. To check how the silanol formation is influenced by atomic strain energy of local sites on the surface, we prepared an amorphous SiO_2 slab with

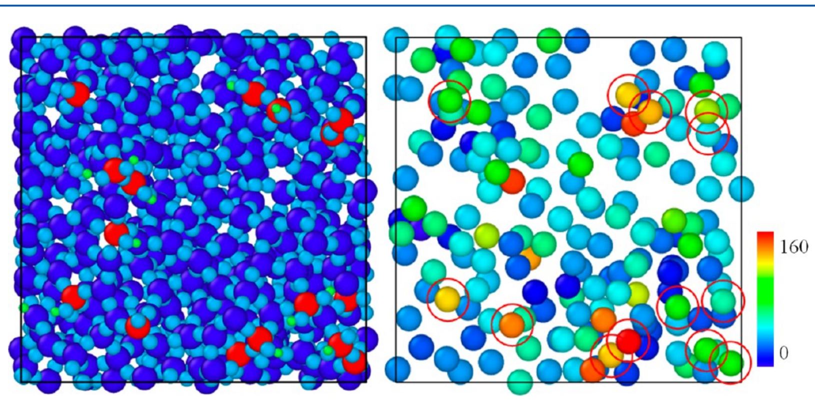

Figure 13. Vertical view of amorphous SiO_2 surface after 300 K simulation. Red atoms are Si atoms of hydroxyls (Left). Strain energy view of surface Si atoms after 300 K hydroxylation simulation. (Right) Red circles represents the position of silanols. Si atoms with relatively high strain energy are generally become the Si of surface hydroxyl.

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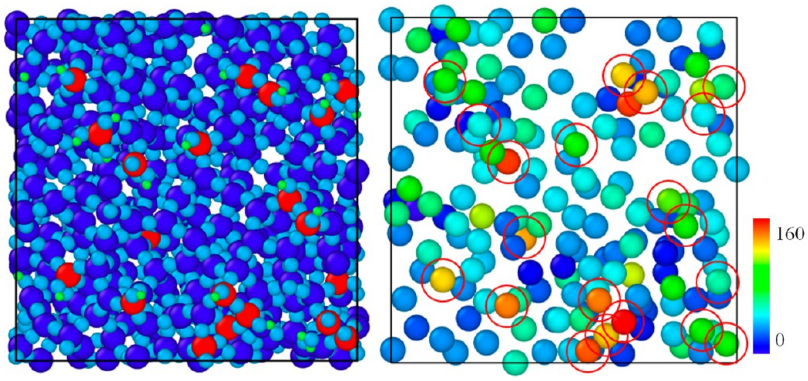

Figure 14. Vertical view of amorphous SiO_2 surface after 500 K simulation. Red atoms are Si atoms of hydroxyls (Left). Strain energy view of surface Si atoms after 500 K hydroxylation simulation. (Right) Red circles represents the position of silanols. Number of hydroxyls are increased when compared to 300 K case, and all of them follows the position of relatively high strained surface Si atoms.

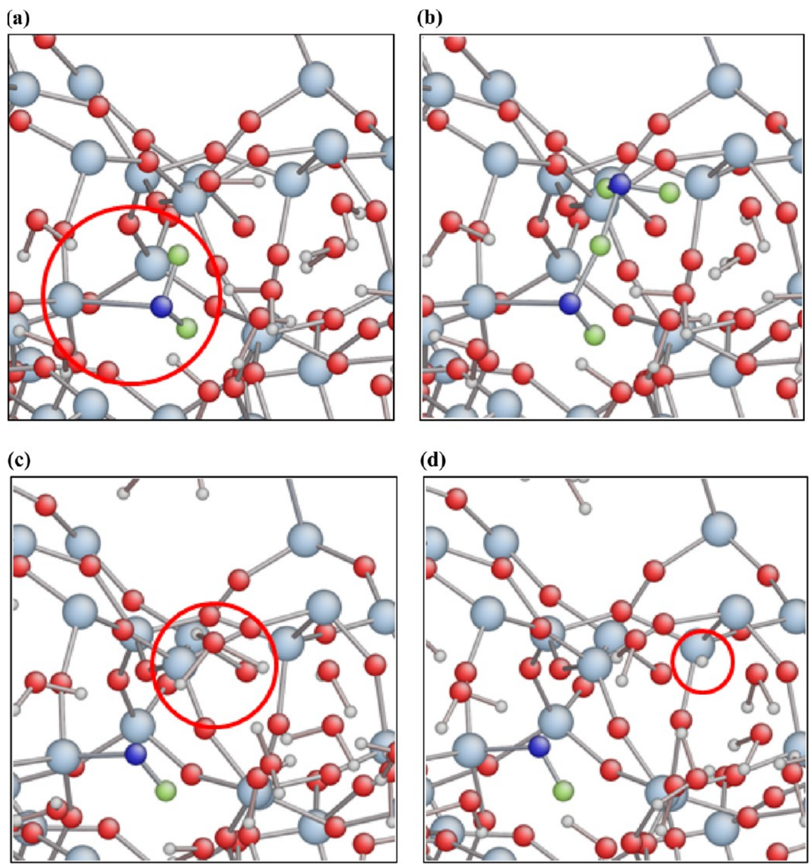

Figure 15. (a) 26.57 ps after simulation. Adsorbed water on a strained silicon structure. (Red circle) Bright gray shows Si, red atoms are O, white atoms are H, blue atoms are O of water, and green atoms are H of water. (b) 26.66 ps after simulation. Donation of proton to the nearby water molecule. (Red circle). (c) 26.665 ps after simulation. Hydronium formation. (Red circle). (d) 26.95 ps after simulation. Proton hopping to other water molecule. (Red circle).

MD simulation. Detailed explanations behind the preparation of a-SiO_2 slab are described in the Figures S2−S4 and Table S1.

To evaluate the reactivity of this amorphous silica surface to water, 200 water molecules with a density of 0.204 kg/dm^3 were deployed between two surfaces of the amorphous silica in random positions, thus forming a water vapor layer of 15 Å thickness. The initial minimum distance among the molecules was maintained at 2.5 Å to prevent any overlapping. Then, the system was exposed to constant temperatures of 300 K and 500 K with an NVT ensemble having 100 fs of damping constant. After 200 ps exposure to water molecules, we observed the formation of surface hydroxyl. The time variation of the hydroxyl number density for the simulation at 300 and 500 K is shown in Figure 12. Both simulations reached a steady state after 100 ps (300 K) and 160 ps (500 K).

Similar to the simulation of silica nano wire with water, hydroxylation of a-SiO_2 slab also favored atomically strained sites on the surface. Figures 13 and 14 show the relationship between silanol formation site and highly strained region of the surface. According to the results from both figures, the hydrolysis reaction clearly prefers the undercoordinated silicon atoms on the surface. The location of red-circled Si atoms of

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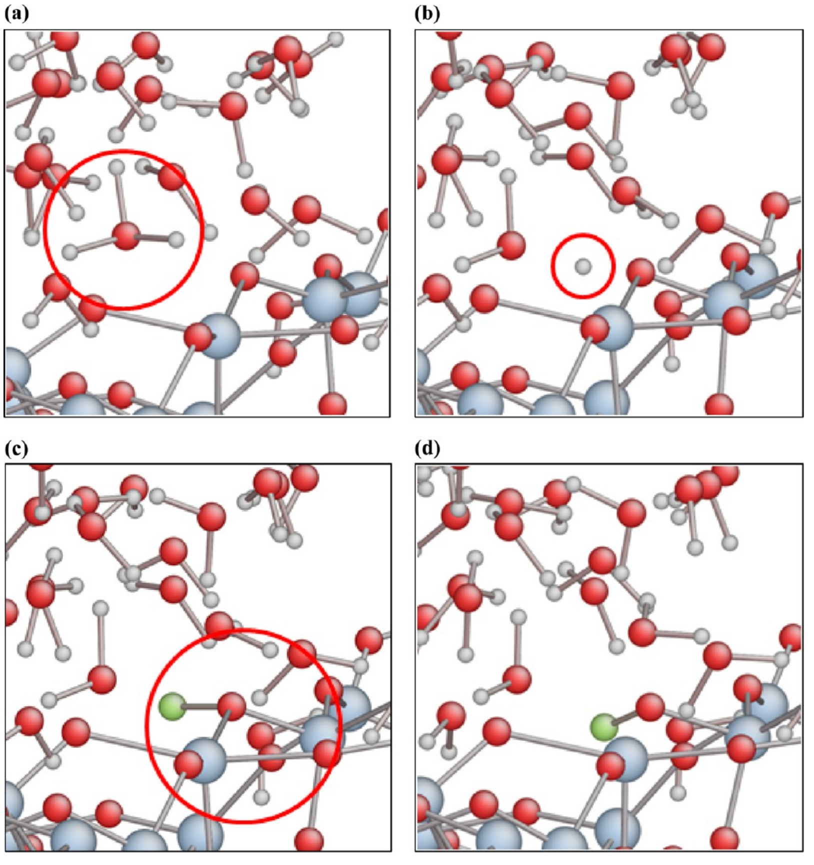

Figure 16. (a) 9.83 ps after simulation. Hydronium inside the water phase. Repeated color scheme with Figure 15. (b) 9.875 ps after simulation. Proton donation from hydronium to a high-strained surface bridge oxygen. (c) 9.88 ps after simulation. Unstable Si—OH—Si structure. (d) 9.925 ps after simulation. One of Si—O bond breaks, forming single silanol on the surface.

Figures 13 and 14 indicates the relatively strained atoms than other atoms, and it matches well with hydroxyl formation sites. We observed that not only the undercoordinated atoms but also the fully coordinated atoms with high local strain energy are responsible for hydroxyl formation.

Under 300 and 500 K conditions, hydroxylation pathway on the a-SiO_2 slab was also similar to silica nano wire case: H_3O^+ ions participated in the hydrolysis reaction. Figure 15 shows the formation of silanol and hydronium molecules on the surface of the amorphous SiO_2 slab, and proton donation from the hydronium molecules by the hopping process. The proton ions in our simulation approximately survive less than 100 fs of time, according to averaged lifetime of all protons during our simulations. The most probable destination of hydrogen ion was a neighboring water molecule forming another hydronium, thus experiencing a proton hopping process. At 26.57 ps (Figure 15(b)), hydroxylation starts from the adsorbed water molecule. At 26.66 ps (Figure 15), adsorbed water molecule donates its proton ion to the nearby water molecule. At 26.665 ps (Figure 15(c)), hydronium ion is formed. At 26.95 ps (Figure 15(d)), hydronium is dissociated and one H atom migrates to the next water molecule, closely placed by hydrogen bond network. After successive hopping, proton forms another silanol, forming hydroxyl on the surface.

In addition, we observed that the donation of a proton to the bridge oxygen atom at surface is also related to the strain energy of the oxygen atom. Due to prolonged Si—O bonds on the amorphous surface, related Si and O atoms on such sites have higher strain energy, which creates a favorable environment for water and hydronium to donate their protons. Figure 16 shows the relationships between the hydroxyl formation site and the strained oxygen atom contour of the surface. At 9.83 ps (Figure 16(a)), hydronium approached close to the surface. At 9.875 ps (Figure 16(b)), proton migrated to the strained Si—O—Si structure on the surface from H_3O^+ ion. At 9.88 ps (Figure 16(c)), Si—OH—Si structure was formed. On the basis of our observation, Si—OH—Si structure was very unstable, so that its survival time was within ~50 fs maximum. Subsequently, at 9.925 ps after simulation (Figure 16(d)), this structure was dissociated into a single silanol and an undercoordinated Si atom, which becomes a new favorable site for additional hydroxyl formation on the surface.

3. CONCLUSIONS

We performed ReaxFF-based molecular dynamics (MD) simulations for hydrolysis reactions on the SiO_2/water interface, using both silica nano wire and an amorphous silica slab. First, we provided additional data for the validation of the ReaxFF SiO_2/water model, by comparison with experimental results and DFT data. Specifically, the energy barrier for the hydroxylation for both strained silica dimer and nonstrained silica dimer is added to the training set, resulting in more reliable description for the hydroxylation reaction. After retraining, both ReaxFF and DFT showed 20 and 30 kcal/mol of energy barrier for the hydroxylation on the strained and nonstrained silica dimer, respectively.

These results indicate that the hydroxylation reaction is more favorable for strained silica compared to that at a nonstrained

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local environment. We observed that a highly strained edge area is preferentially responsible for silanol formation, rather than the lower-strain silicon atoms in the body of the nano wire.

In addition, we detected three specific hydroxylation paths. First, we observed the silanol formation initiated from adsorbed water molecule, forming a hydronium ion by donating its proton to a nearby water molecule, leaving silanol on the surface, and forming—usually solvated—H_3O^+ as a product. The second path for silanol formation was the dissociation of hydronium ion. A H_3O^+, formed in the first method of silanol formation, donates one of H ions to the bridge oxygen on the surface, creating silanol on the donated spot, and leaving a water molecule as a product. The third method of silanol formation was the direct dissociation of an adsorbed water molecule. After adsorption, one of the H of Si—H_2O moves to bridge oxygen, forming two silanols at once.

We observed a strong relationship between the temperature of system, the reaction pathway for the hydroxylation, and the existence of water clusters. Our result implies that high temperatures (>1200K) induced the dissociation of hydrogen bond network, as well as water cluster structures. This indicates that the reduced number of hydronium ions, and direct dissociation of adsorbed water molecule become major hydrolysis pathway at such temperature. Meanwhile, from 500 K to 1000 K of the temperature range, the existence of water dimer and nearby water molecules played important role on hydrolysis reaction involving H_3O^+ ions. Due to water clusters, water molecules easily find the target—nearby water molecules—for the donation or adsorption of proton ions, inducing the formation and dissociation of hydronium ions. This may suggest the existence of optimum temperature for water−Si interactions.

To investigate the trends of silanol formation on a larger silica structure, an amorphous SiO_2 slab was prepared by ReaxFF simulation. Bulk properties of the surface and slabs were compared to and validated with the experimental results. MD simulations on the interaction of water molecules with the amorphous silica slab confirm that the hydroxyl groups on the silica surface favor the locally strained SiO_2 structure and undercoordinated silicon atoms. The comparison between the surface strain energy map and silanol formation map supports this phenomenon; the silanol formation sites match with the locations of the more highly strained Si atoms. We also observed proton hopping events among the water molecules, producing reactive hydronium molecules, which can donate a proton to the surface oxygen atoms. Highly strained oxygen atoms and undercoordinated oxygen atoms can also be capable of receiving a proton from water. These observations agree with those from the simulation using the silica nano wire, as well as previous studies reported by other researchers.[22]

■ ASSOCIATED CONTENT

Ⓢ Supporting Information

The Supporting Information is available free of charge on the ACS Publications website at DOI: 10.1021/acs.jpcc.5b09784.

> Tracking of number of hydroxylation data, and brief explanation about annealing of quartz, and formation of amorphous SiO_2 slab (PDF)

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Notes

The authors declare no competing financial interest.

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